

RESEARCH

Open Access



A temporal-spectral graph convolutional neural network model for EEG emotion recognition within and across subjects

Rui Li¹, Xuanwen Yang¹, Jun Lou¹ and Junsong Zhang^{2*}

Abstract

EEG-based emotion recognition uses high-level information from neural activities to predict emotional responses in subjects. However, this information is sparsely distributed in frequency, time, and spatial domains and varied across subjects. To address these challenges in emotion recognition, we propose a novel neural network model named Temporal-Spectral Graph Convolutional Network (TSGCN). To capture high-level information distributed in time, spatial, and frequency domains, TSGCN considers both neural oscillation changes in different time windows and topological structures between different brain regions. Specifically, a Minimum Category Confusion (MCC) loss is used in TSGCN to reduce the inconsistencies between subjective ratings and predefined labels. In addition, to improve the generalization of TSGCN on cross-subject variation, we propose Deep and Shallow feature Dynamic Adversarial Learning (DSDAL) to calculate the distance between the source domain and the target domain. Extensive experiments were conducted on public datasets to demonstrate that TSGCN outperforms state-of-the-art methods in EEG-based emotion recognition. Ablation studies show that the mixed neural networks and our proposed methods in TSGCN significantly contribute to its high performance and robustness. Detailed investigations further provide the effectiveness of TSGCN in addressing the challenges in emotion recognition.

Keywords EEG signal, Affective computing, Graph neural network, Cross-subjects

1 Introduction

1.1 Affective computing and emotion recognition

Affective computing is an interdisciplinary research area involving computer science, psychology, and neuroscience [1], whose ultimate goal is to build computers with human-like emotions that can recognize, interpret, and simulate those emotions. Affective computing is applied widely in various fields, such as health monitoring [2],

online social interaction [3], and brain-computer interaction [4]. One of the most important research areas in affective computing is emotion recognition, which is fundamental for computers to recognize human emotions. Emotion recognition leverages computers to build emotional systems and detect human emotional states.

The first step in building an emotional system is to define the emotion category. Psychologists have built two models (discrete emotion and dimensional emotion) to describe the emotion. The discrete emotion model consists of fixed emotional words, such as those in Plutchik's wheel of emotion (anger, fear, sadness, disgust, surprise, expectation, trust, and joy) [5]. The dimensional emotion model is a continuous multidimensional representation of emotions, such as valence, arousal, and dominance [6]. Valence indicates the degree of pleasure of emotion, arousal indicates the intensity of emotion, and dominance

*Correspondence:

Junsong Zhang
zhangjs@xmu.edu.cn

¹ Brain Cognition and Computing Lab, National Engineering Research Center for E-Learning, Faculty of Artificial Intelligence in Education, Central China Normal University, Wuhan, Hubei, China

² Brain Cognition and Intelligent Computing Lab, Department of Artificial Intelligence, School of Informatics, Xiamen University, Xiamen, Fujian, China

indicates the autonomy of emotion. These two models, which define emotion from different aspects, are applied pervasively to distinguish different kinds of emotions in psychology and neuroscience.

To detect human emotional states, various data, including non-physiological and physiological signals, are recorded. Non-physiological signals, which mainly arise from human behavior, such as facial expression, human gestures, texts, and voice, are feasible to collect but easily influenced by variation in illumination, target partial occlusion, and deformation. Besides, humans may intentionally hide their emotions, sometimes making non-physiological signals less conceivable [7, 8]. At this point, the method of emotion recognition based on physiological signals demonstrates more significant potential because internal neural changes cannot be intentionally hidden or controlled [9]. Physiological signals, such as the Electroencephalogram (EEG), Magnetoencephalogram, functional Near Infrared Spectroscopy, and functional Magnetic Resonance Imaging (fMRI), detect neural activity from the human brain without invasive surgery. Data recorded from these techniques reflect the brain activity of emotional recognition and genuinely capture the inner response to emotional stimuli. Among these techniques, EEG is widely applied in emotion detection due to its advantages of high temporal resolution, relatively low price, and portability [10].

However, there are several challenges to building a robust emotion recognition system based on EEG signals. One challenge is extracting high-level feature patterns, which could accurately reflect different emotions from EEG signals. Scalp EEG signals are often subject to a low signal-to-noise ratio and artifacts, like ocular artifacts, muscle movements, head movements, etc. [11]. Besides, according to the neural mechanism of emotion cognition, high-level features related to emotion are distributed sparsely in three domains: space, time, and frequency. In the spatial domain, multiple channels across the brain scalp record neural activity from brain regions processing emotion [12]. In the time domain, cognitive processing of emotional stimuli occurs in different time windows [13]. In the frequency domain, emotional responses are achieved by communicating between brain regions through neural oscillation in specific frequency bands [14]. Thus, distinct patterns of neural activity in the three domains are essential to forming robust high-level features for emotion recognition.

Another challenge in emotion recognition is the difference between emotional responses across subjects. Variations toward emotional stimuli across subjects are caused mainly by subjective ratings and mismatched domains. Subjective rating, which indicates a person's cognitive reappraisal of emotional stimulus [15], is related to

personal cultural background, experience, and traits [16]. Consider a movie scene in which one audience moves while leaving another unaffected. The differences in subjective ratings lead to high individual variability in emotion evaluation, which causes inconsistencies between predefined labels and emotions elicited by individuals. A mismatched domain means the feature distributions of emotional data across subjects are not identical [17, 18]. Most previous studies focused on emotion recognition, where training and testing data came from the same domain. Under this assumption, the feature distribution of training data is identical to that of testing data. The learning model from the source domain data is used directly to predict the label of the target domain data [19]. However, in real situations, the feature distribution of data across different subjects violates this assumption, which leads to a decreased generalization for the learning model when it learns features from one subject to predict the labels of another. Thus, to solve the challenge of cross-subject emotion recognition, both subjective ratings and mismatched domains must be considered.

1.2 EEG-based emotion recognition

In recent years, computational algorithms for emotion recognition have extended from traditional machine learning to deep learning to extract robust features from EEG signals. Compared with models in traditional machine learning, such as Support Vector Machine (SVM), K-Nearest Neighbor, etc., deep neural networks in deep learning automatically extract representative features from original EEG signals without much expert knowledge [20]. Thus, more and more researchers have adopted deep learning for emotion recognition. Several deep learning models have been developed to extract robust features from EEG signals from spatial, time, and frequency domains.

Previous studies mainly focused on extracting features in the spatial or time domain of EEG signals. Researchers have developed various Recurrent Neural Networks (RNNs) to extract temporal information from EEG signals. For example, to recognize emotion, Alhagry et al. used LSTM to extract temporal features from EEG signals [21]. Ma et al. proposed a residual Long Short-Term Memory (LSTM) model that shares time weights among multiple modalities for emotion recognition [22]. Xiao et al. integrated the temporal attention mechanism into a bi-directional LSTM to explore temporal dependencies of the four-dimensional EEG signal representations, which can extract the temporal dependencies and explore the importance of different temporal slices [23]. The commonly used method to extract spatial information is to build a feature map from EEG signals and use Convolutional Neural Network (CNN) to extract

high-level representation from the feature map. Wen et al. used the Pearson Correlation Coefficients to reconstruct multi-channel EEG signals, which were then used as CNN input to obtain spatial information for emotion recognition [24]. Li et al. used Diffusion Entropy (DE) features and electrode positions to generate 2-D maps [14]. Siddharth et al. used Power Spectral Density (PSD) in three frequency bands to compose an RGB heat-map image [25]. Then, they used CNNs to capture spatial information from these feature maps to recognize a subject's emotions. Moon et al. proposed a CNN model combined with brain connectivity features to explain the synchronous activation of emotional responses between different brain regions [26]. However, features in a single domain cannot fully represent emotional responses in EEG signals.

The mentioned methods overlook the complementary information of spatiotemporal connections in the brain, which, to some extent, limits the classification capability of the model. Therefore, some research has begun to combine multiple feature extraction methods to achieve enhanced results. Li et al. extracted temporal, spatial, and connectivity features from EEG signals for emotion recognition [27]. Yang et al. proposed a Parallel Convolutional Recurrent Neural Network (PCRNN) that combines CNNs with LSTM [28]. In PCRNN, sequential EEG signals are converted into a sequence of 2D-like frames, and a CNNs module is used to mine the inter-channel correlation between adjacent channels. Then, the LSTM module is used to mine the temporal information from EEG signals. Zhang et al. proposed a Spatial–Temporal Recurrent Neural Network, which contains a multi-directional RNN and a bidirectional RNN [29]. The multi-directional RNN captures remote spatial information by traversing the spatial region of each time slice in different directions. The bidirectional RNN learns discriminative features that describe the time dependence of human emotional responses. Although these studies utilized CNNs to learn spatial information from adjacent channels, extracting information between remote channels isn't easy. In other words, CNNs are unsuitable for extracting sparsely distributed information in the spatial domain.

Graph Neural Network (GNN), built under graph theory to capture relations between adjacent and remote neighbors, solves the problem well. In GNN, weights of the gradual connections between channels are set according to prior knowledge from brain anatomical or functional networks. The graph representation method has recently been applied in many fields [30, 31]. Song et al. proposed a Dynamic Graph Convolutional Neural Network (DGCNN) model to recognize emotion [32]. In DGCNN, a learnable adjacency matrix, which learns

the connectivity between each vertex node of the graph, is used to learn the internal relationship between EEG channels. Pan et al. proposed a Spatio-Temporal Self-Constructing Graph Neural Network (ST-SCGNN) for cross-subject emotion recognition and consciousness detection [33]. ST-SCGNN combined the connection pattern features and activation pattern features, enhancing cross-subject emotion recognition effectiveness in a complementary manner. Zhong et al. developed a Regularized Graph Neural Network to capture local and global relationships between EEG channels [34]. In addition, Song and his colleagues proposed a Variational Instance Adaptive Graph approach to reduce both individual differences and dynamic uncertainty relationships between different EEG channels [35]. Despite the powerful performance of GNN in extracting sparsely distributed information from the EEG spatial domain, it is difficult for GNN to extract contextual information in the time domain. Thus, it is still challenging to integrate multi-dimensional EEG features more effectively and capture sparsely distributed information.

1.3 Learning with noisy labels

Subjective rating causes inconsistency between predefined labels and actual emotional responses, which can be regarded as noisy labels in a predefined label set. To solve the learning problem caused by noisy labels, previous studies commonly developed new approaches based on three key ingredients: data, objective function, and optimization strategy [36].

Data For data, the key is to discover a potential noise transfer matrix, which is essential for analyzing patterns from true labels to noisy labels. The noise transfer matrix specifies the transition probability from each true label to each noisy label for the objective function. In deep learning, the most straightforward learning with noisy labels is to add an adaptation layer to estimate the probability transition matrix or learn through noise transfer moments [37]. That is, the training of noisy labels is approximately equal to clean labels through loss correction. Patrini et al. used a noise transfer matrix to correct the loss function in forward and backward propagation for noisy labels [38].

Objective function For objective function, the key is to design a robust and noise-resistant loss function. For example, Azadi et al. proposed an auxiliary regularization to select reliable labels from noisy labels by exploiting the mutual context information in training data [39]. Berthelot and his colleagues introduced an explicit regularization, Minimum Entropy Regularization, in the MixMatch model for semi-supervised learning [40]. Zhang et al. proposed a generalized cross-entropy loss function that

considered both Categorical Cross Entropy (CCE) loss and Mean Absolute Error loss [41].

Optimization strategy For optimization strategy, the key is to dynamically optimize the exploring progress, such as MentorNet proposed by Jiang et al. and co-teaching proposed by Han et al. [42, 43]. In Jiang's work, the MentorNet is proposed to dynamically learn a data-driven sample weight scheme that makes StudentNet focus on correct labels. In Han's work, co-teaching is two deep neural networks that simultaneously select data with possibly clean labels and communicate with each other to update each network dynamically.

Although these studies have proposed new methods to address the challenge of learning with noisy labels, they did not focus on the subjective and objective ratings in affective computing. How to address the consistency between subjective rating and predefined labels in emotion recognition is still a challenge.

1.4 Transfer learning on emotion recognition

Transfer learning mainly aims to transfer knowledge between training and test data while relaxing the assumption that these two data are independent and identically distributed. Transfer learning uses knowledge in the source domain to help learn in the target domain; thus, transfer learning is very suitable for dealing with mismatched domains across subjects in Affective computing. Quan et al. proposed a domain transfer learning model incorporating substructure selection, multi-source domain selection, and subdomain adaptation from various sources for emotion classification [44].

There are two main kinds of transfer learning methods. The first kind is based on statistical moment matching methods, including Maximum Mean Discrepancy (MMD) and CORrelation ALignment [45–47]. Li et al. used a Deep Adaptation Network to improve the accuracy of cross-subject emotion recognition based on EEG signals [48]. Zhang et al. used MMD to measure the similarity between individuals [49]. Then, they extracted pre-existing EEG data of similar individuals and used the TrAdaBoost algorithm to build a user-dependent model for the target individual.

The second kind is based on the adversarial loss method. Borrowing the idea from Generative Adversarial Network, this method, Domain Adversarial Neural Network (DANN), uses a discriminator to distinguish whether extracted features are from the source domain or target domain [50]. Inspired by the lateralization of brain neural activity during emotional cognition, Li et al. proposed a Bi-hemispheres Domain Adversarial Neural Network model (BiDANN), which combines asymmetry between left and right hemispheres with

the DANN confrontation mechanism to recognize emotional responses from different subjects [51].

Considering these challenges in emotion recognition, we propose a novel neural network: Temporal-Spectral Graph Convolutional Neural Network (TSGCN). TSGCN not only extracts robust feature patterns from EEG signals in multiple domains but also minimizes the differences in emotional responses across subjects.

First, to extract robust feature patterns from EEG signals, TSGCN uses a Bi-directional Long Short-Term Memory network (BiLSTM) to extract temporal-spectral features from the time–frequency domains. Then, TSGCN regards discrete EEG channels across the brain scalp as nodes in the graph and uses a Dynamic Graph Convolutional Neural Network (DGCNN) to capture topological relationships between EEG channels in the spatial domain. Specifically, to fuse multi-dimensional features in time-spatial-frequency domains and form a high-level representation of emotional response, TSGCN retains temporal information while passing the features from BiLSTM to DGCNN.

Second, to minimize the individual difference in response to emotional stimuli, the subjective rating problem is formalized as noisy tags, which are fuzzy classes with emotional confusion. Then, to reduce the influence of noisy tags on prediction results, a Minimum Category Confusion (MCC) loss function is added in TSGCN. In addition, to solve the problem of mismatched domains across subjects and further improve the generalization of TSGCN, we propose a Deep and Shallow feature Dynamic Adversarial Learning (DSDAL) that aligns shallow time–frequency features with deep graph node features in TSGCN. DSDAL uses dynamic factors to evaluate the importance of both shallow and deep features. Then, it aligns the feature distribution of different individuals from the shallow to the deep neural network to improve the accuracy of cross-subject classification.

To validate the performance of the TSGCN model, we conducted extensive experiments on public datasets. Besides, we performed a detailed investigation to verify the effectiveness of TSGCN in addressing the challenges in emotion recognition. The main contributions of this paper are as follows:

- (1) We propose a TSGCN model to recognize emotional responses based on EEG signals. TSGCN integrates the information from spatial, time, and frequency domains to form a high-level representation of emotion. TSGCN captures the discreteness of emotional responses in time–frequency domains and the discreteness of emotional responses from the distributed channels in the spatial domain.

- (2) We add the MCC loss and DSDAL in TSGCN to minimize the influence of individual differences in emotion recognition. MCC reduces the influence of subjective ratings by discriminating the fuzzy classes of emotional responses, while DSDAL improves the generalization ability of TSGCN by aligning the distributions of both time–frequency features and graph node features across subjects.
- (3) We conduct extensive experiments using two public datasets: SEED and DEAP. Experimental results show that the TSGCN model performs better than the most advanced methods in EEG-based emotion recognition.
- (4) We perform a detailed analysis of the challenges in emotion recognition, which include the relationships between channels, the influence of subjective ratings, and the alignment of mismatched domains across subjects. From the results, we glean a better understanding of how these challenges influence emotion recognition accuracy. We present the effectiveness of each module in TSGCN in addressing these challenges.

The remainder of this paper is organized as follows: In Sect. 2, we propose the TSGCN model for EEG-based emotion recognition. In Sect. 3, we validate the performance of TSGCN using two public datasets. In Sect. 4,

we perform a detailed analysis and discuss how TSGCN addresses the challenges in emotion recognition. In Sect. 5, we conclude the paper.

2 Proposed method

We propose a TSGCN model to deal with the problem of cross-subject emotion recognition. Our methodology contains three parts: First, we introduce the framework of TSGCN for emotion recognition. Second, we describe a minimum category confusion loss function, which is used to reduce the influence of subject rating. Third, we introduce details of the deep and shallow feature dynamic adversarial learning for cross-subject problems and the training process of TSGCN.

2.1 Temporal-spectral graph convolution network

TSGCN aims to extract high-level emotional representation from time, frequency, and spatial domains of EEG signals. The framework of TSGCN used for emotion recognition is shown in Fig. 1.

TSGCN uses EEG features in frequency bands as input to capture information in the frequency domain. These features are extracted from EEG signals by using popular methods, including Differential Entropy (DE), Power Spectral Density (PSD), and Wavelet Transform (WT) [58]. Given EEG features, $X \in \mathbb{R}^{n \times t \times d}$, and labels, $Y \in \mathbb{R}^{c \times 1}$, where n denotes the number of electrode

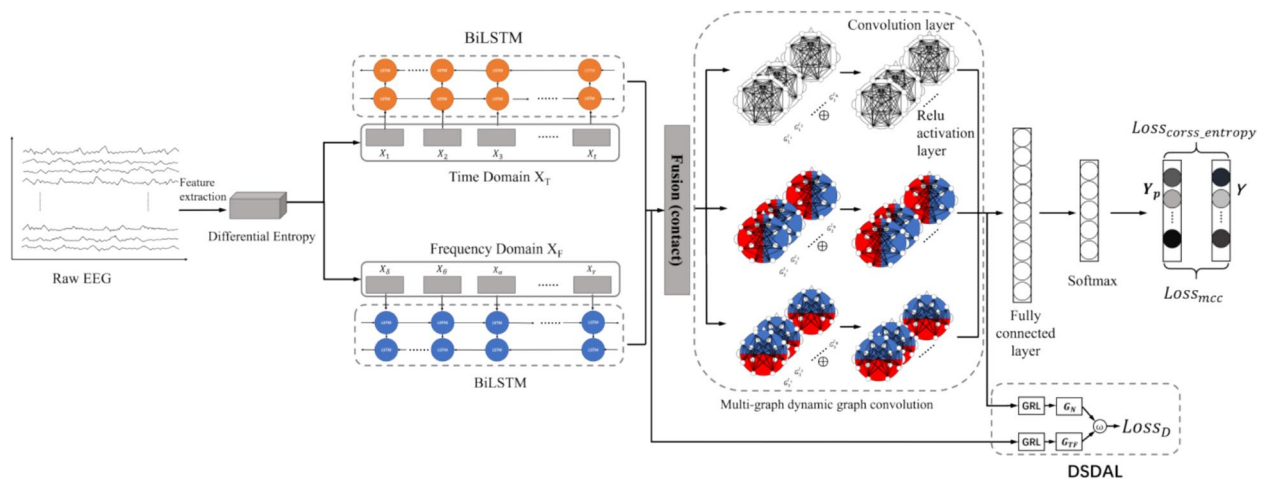


Fig. 1 The Framework of TSGCN. The main frame part of TSGCN consists of two modules, the BiLSTM module, and the multi-graph dynamic graph convolution module, which are indicated by gray dash lines. DSDAL represents the module of Deep and Shallow feature Dynamic Adversarial Learning. The inputs of TSGCN are DE features in frequency bands. DE features are sent to two BiLSTM neural networks to extract information in time and frequency domains, respectively. Then, time–frequency features from BiLSTM are fused and sent to a multi-graph dynamic graph convolution neural network, which is constituted by three parallel dynamic graph convolution neural networks using different connectivity matrices. There is a fully connected layer behind the multi-graph dynamic graph convolution, and the predicted labels are derived using softmax. The prediction loss function and the MCC loss function are derived from estimating the difference between the predicted labels and the true labels. The loss function of DSDAL is derived from both the loss of the shallow discriminator and the loss of the deep discriminator. GRL (Gradient Reversal Layer) helps with domain adaption by reversing the gradient. G_N processes node features in the graph, and G_{TF} processes time–frequency features

channels, t denotes the number of time sequences, d denotes feature dimension, and C denotes the number of classes. For these features, TSGCN first uses an RNN model to extract contextual information in the time domain. Compared with simple RNN and LSTM, BiLSTM connects two unidirectional LSTMs with opposite directions to the same output, which retains the past and future information to better understand contextual information. Thus, TSGCN uses BiLSTM to obtain frequency domain information $X_F \in \mathbb{R}^{n \times d}$ and time domain information $X_T \in \mathbb{R}^{n \times d}$ from EEG features, where d' denotes the output dimension of BiLSTM.

For the output features of BiLSTM, the Dynamic Graph Convolutional Neural Network is used to extract topological information in the spatial domain. A learnable adjacency matrix $A \in \mathbb{R}_{n \times n}$ in DGCNN represents the topological structure between each electrode, where n represents the number of channels of EEG signals. We initialize the learnable adjacency matrix, A , according to Zhong et al. [28] as follows:

$$A_{pq} = \min \left(1, \frac{\delta}{d_{pq}^2} \right) \quad (1)$$

where θ is a calibration constant (we chose $\theta=0.1$), d_{pq} represents the physical distance between channel p and q , where $p, q=1, 2, \dots, n$. Achard and Bullmore observed that sparse fMRI networks, with around 20% of possible connections, typically maximize network efficiency [52]. Thus, we set $\delta=0.1$ so that about 20% of entries in A are non-negligible, considering values larger than 0.1 as such connections.

Due to the variation of topological information in the brain network and the asymmetry of functionality in the left and right brain hemispheres, we used three connection

matrices A_{all} , A_{lr} , and A_{fb} . To reconstruct brain channel information, brain network A_{all} contains the connectivity information between all electrode channels, brain network A_{lr} contains the connectivity information between channels in the left and right hemispheres, and brain network A_{fb} contains the connectivity information between channels in the up and down hemispheres. The schematic diagram of left-right connectivity and up-down connectivity is shown in Fig. 2.

To retain the contextual information in the time domain, while TSGCN passes the features from BiLSTM to DGCNN, we fuse time-frequency information into three spatial information matrices as follows:

$$G_k^F = \sigma \left(\hat{A}_k X_F W \right) \quad (2)$$

$$G_k^T = \sigma \left(\hat{A}_k X_T W \right) \quad (3)$$

where $k \in [1,3]$ denotes the left and right, up and down, and the entire connection, respectively. σ is a non-linear activation function, we use the Relu function, $\hat{A}_k = D^{-1/2} A_k D^{-1/2}$ represents the normalization of the learnable connection matrix, and D is the degree matrix of A_k , A_k is the connection matrix in the set, $W \in \mathbb{R}^{d \times d}$. Dynamic graph convolution neural networks in the frequency domain share a set of parameters G_k^F and in the time domain, these neural networks share a set of parameters G_k^T .

Splicing the information of different connected graph nodes, the formula is as follows:

$$G = |_{k \in [1,3]} \left[G_k^F W_1 + b_1 || G_k^T W_2 + b_2 \right] \quad (4)$$

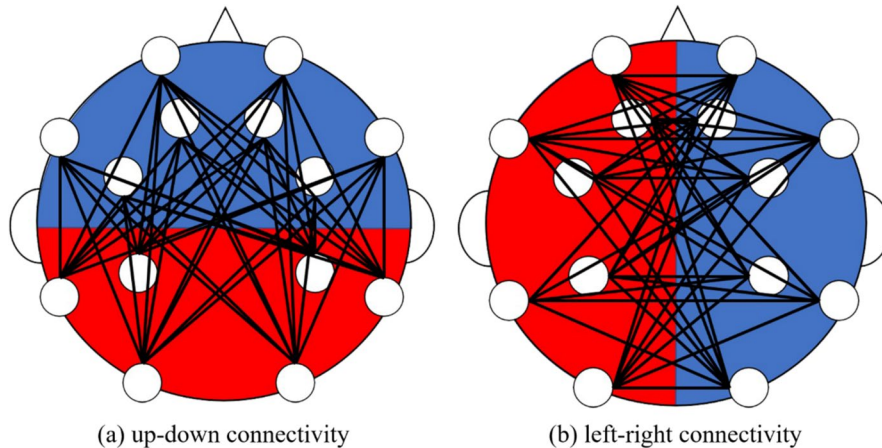


Fig. 2 The channel division in the brain. **(a)** The channels are divided according to the upper and lower brain regions. **(b)** The channels are divided according to the left and right brain regions. Black lines indicate the connections between channels in the two brain regions

where $\parallel$ represents the concatenation of different connection methods, $\parallel$ represents the concatenation of frequency domain and time domain features. G_k^F represents the frequency feature after graph convolution, and G_k^T represents the time domain features after graph convolution, k is represented as different connections mentioned above, $W_1 \in \mathbb{R}^{d'' \times d''}$ and $W_2 \in \mathbb{R}^{d'' \times d''}$ are represented as weight matrix, $b_1 \in \mathbb{R}^{n \times d''}$ and $b_2 \in \mathbb{R}^{n \times d''}$ are represented as bias matrices.

After the graph is convolved, the node features are aggregated across all nodes. Finally, behind the network is a fully connected layer, which uses a Softmax function for prediction, according to the following formula:

$$Y_p = \text{Softmax}(\text{Relu}(\text{AGG}(G)W_3 + b_3)) \quad (5)$$

where Y_p represents the predicted label, $W_3 \in \mathbb{R}^{d'' \times d''}$ represents the output weight matrix, $\text{AGG}(\cdot)$ represents node aggregation, $b_3 \in \mathbb{R}^{N \times d''}$, N is the number of samples.

During the training process, TSGCN uses a cross-entropy loss function to estimate the difference between predicted labels and true labels. The loss function is expressed as follows:

$$\text{Loss} = \text{corss_entropy}(Y, Y_p) + \alpha \|\theta\| \quad (6)$$

where θ denotes all parameters of the model and α is the regularization weight.

2.2 Minimum category confusion

A subjective rating may not match a predefined emotion label, resulting in poor model performance. We define the concept of confusion as emotional responses that are misjudged and formulize it as a loss function, which guides the misjudgment of emotional confusion during the training process [53]. The category correlation between two classes j and j' of the i -th instance is defined as follows:

$$\hat{y}_{ij} = \frac{\exp(Z_{ij})}{\sum_{j=1}^C \exp(Z_{ij})} \quad (7)$$

$$C_{jj'} = \hat{y}_j^T \hat{y}_{j'} \quad (8)$$

where Z_{ij} represents the j -th class output of the i -th instance of the classifier, and $\hat{Y}_{ij}$ is the predicted probability that the i -th instance belongs to class j , after normalization through the Softmax function, it ensures that the sum of probabilities across all classes is 1, C represents the number of classes, $\hat{y}_j^T$ represents the transpose of the probability matrix of j -th class in the instance, $\hat{y}_{j'}$ represents the probability matrix of j' -th class in the instance, and $C_{jj'}$ represents the correlation between class j and

class j' , computed as the dot product of the probability vectors of the two classes.

In the category of emotion confusion, the distribution of predicted labels may be imbalanced in multiple emotion categories, which is not conducive to our final classification. Due to the varying amount of information provided by samples of different categories, their contributions to class correlation are also different. Therefore, we add a weighting mechanism based on instance output, hoping that instances consistent with sentiment labels will be more important in category confusion. The weight of i -th sample is defined as follows:

$$H(\hat{Y}_i) = - \sum_{j=1}^C \hat{y}_{ij} \log \hat{y}_{ij} \quad (9)$$

$$\text{Weight}_i = \frac{\exp(-H(\hat{Y}_i))}{\sum_{i=1}^N \exp(-H(\hat{Y}_i))} \quad (10)$$

where $H(\hat{Y}_i)$ denotes the information entropy of i -th instance, Weight_i denotes the weight of the i -th sample to category confusion modeling, and N denotes the number of samples.

Therefore, the weighted category confusion matrix is defined as:

$$C_{jj'} = \hat{y}_j^T W \hat{y}_{j'} \quad (11)$$

In the derivation above, $C_{jj'}$ is a good measure of the confusion of each class between j and j' . Therefore, we only need to minimize the loss of confusion across categories. $j \neq j'$ means that all samples are correctly classified into the right categories. Therefore, the final Minimum Category Confusion (MCC) loss is defined as follows:

$$C_{jj'} = \frac{C_{jj'}}{\sum_{j'=1}^C C_{jj'}} \quad (12)$$

$$\text{Loss}_{\text{MCC}}(Y_p) = \frac{1}{C} \sum_{j=1}^C \sum_{j' \neq j}^C \tilde{C}_{jj'} \quad (13)$$

where $\tilde{C}_{jj'}$ denotes the normalization of the category confusion matrix.

2.3 Deep and shallow feature dynamic adversarial learning

The main difficulty of cross-subject emotion recognition is that there are certain differences in the feature distribution of EEG signals between each subject. Thus, the feature distribution extracted from a model is different across subjects. To reduce this difference in TSGCN, we propose Deep and Shallow feature Dynamic Adversarial

Learning (DSDAL) based on DAAN [54]. Compared with previous studies, DSDAL in our work considers the feature distribution in both the time–frequency domain and fine-grained information at the nodes level. Importantly, the alignment of feature distribution in both the time–frequency domain and nodes level are evaluated dynamically during training to achieve optimization between them.

Given a source domain, $D_S = (x_s^r; y_s^r)$, and a target domain, $D_T = (x_T^r; y_T^r)$, D_S and D_T obey different probability distributions p and q , respectively, where $p \neq q$. The loss of domain classifier is defined as follows:

$$\Phi D = - \sum_{r=1}^N (\log(p(0|X_S^r, \theta_D))) + (\log(q(1|X_T^r, \theta_D))) \quad (14)$$

where $p(\cdot)$ denotes domain probabilities of time–frequency feature, θ_D denotes the parameters of the domain classifier, and N is the number of samples. ΦD is the cross-entropy loss of the domain discriminator. The formula is designed to align the marginal distribution between the source domain and the target domain, thereby reducing the differences in the feature distribution of EEG signals across subjects in emotion recognition.

The shallow discriminator aims to align the time–frequency feature distribution between source and target domains. Therefore, the loss of shallow discriminator is as follows:

$$Loss_F = - \sum_{i=1}^N (\log(p(0|X_{FS}^r, \theta_F))) + (\log(q(1|X_{FT}^r, \theta_F))) \quad (15)$$

$$Loss_T = - \sum_{i=1}^N (\log(p(0|X_{TS}^r, \theta_T))) + (\log(q(1|X_{TT}^r, \theta_T))) \quad (16)$$

where X_{FS}^r denotes the frequency feature of the source domain, X_{FT}^r denotes the frequency feature of the target domain, X_{TS}^r denotes the time feature of the source domain, X_{TT}^r denotes the time feature of the target domain, r represents the r -th sample of feature, θ_F denotes the parameters of frequency domain classifier, θ_T denotes the parameters of time domain classifier.

$$p(0|X_{FS}^r; \theta_F) = \text{softmax}_0 (\sigma(X_{FS}^r) W_F) \quad (17)$$

$$p(1|X_{FT}^r; \theta_F) = \text{softmax}_1 (\sigma(X_{FT}^r) W_F) \quad (18)$$

$$p(0|X_{TS}^r; \theta_T) = \text{softmax}_0 (\sigma(X_{TS}^r) W_T) \quad (19)$$

$$p(1|X_{TT}^r; \theta_T) = \text{softmax}_1 (\sigma(X_{TT}^r) W_T) \quad (20)$$

where σ is the nonlinear activation function, the matrix parameters in the frequency and time domain classifier are represented by $W_F \in \mathbb{R}^{d \times 2}$ and $W_T \in \mathbb{R}^{d \times 2}$, respectively.

The deep discriminator aims to align feature distributions in graph nodes between source and target domains. In the dynamic adversarial learning of deep and shallow features, the deep discriminator draws on the loss of the node discriminator of the Regular Graph Neural Network (RGNN) and is calculated as follows:

$$Loss_N = - \sum_{r=1}^N \sum_{j=1}^n \log(p_j(0|G_{NS}^r, \theta_G)) + \log(p_j(1|G_{NT}^r, \theta_G)) \quad (21)$$

where $p_j(\cdot)$ denotes domain probabilities of graph node, n denotes the number of nodes, G_{NS}^r denotes the graph node information of the source domain, G_{NT}^r denotes the graph node information of the target domain, r represents the i -th sample of feature, θ_G denotes the parameters of node domain classifier.

$$p(0|G_{NS}^r; \theta_G) = \text{softmax}_0 (\sigma(G_{NS}^r) W_G) \quad (22)$$

$$p(1|G_{NT}^r; \theta_G) = \text{softmax}_1 (\sigma(G_{NT}^r) W_G) \quad (23)$$

where σ is the nonlinear activation function, $W_G \in \mathbb{R}^{d \times 2}$ denotes the matrix parameter in the node domain classifier.

We adopt the same approach as DAAN by using dynamic adversarial factors to evaluate the distribution of time–frequency features and graph node features. We denote the distance of feature distribution as loss; thus, the dynamic adversarial factor is as follows:

$$\omega = \frac{Loss_N}{Loss_N + \frac{1}{2}(Loss_F + Loss_T)} \quad (24)$$

Therefore, the loss of dynamic adversarial learning of deep and shallow features is as follows:

$$Loss_D = \omega Loss_N + (1 - \omega) \left(\frac{Loss_F + Loss_T}{2} \right) \quad (25)$$

The final loss function is defined as follows:

$$Loss' = Loss + Loss_{MCC} + Loss_D \quad (26)$$

See Algorithm 1 for detailed procedures on training TSGCN.

Algorithm 1 Procedures for training optimal TSGCN model

Input: Train data X_S and labels Y , test data X_T , learning rate η , number of epochs m , other regularization hyper-parameters;
Output: The learned model parameters in TSGCN;
Initialize model parameters;
Initialize learnable adjacency matrix A_{all}, A_{lr}, A_{fb} ;
For $i = 1:m$ do
 Calculate X_F and X_T on the BiLSTM
 Calculate G_F and G_T on the Dynamic Graph Convolutional Neural Network with A_{all}, A_{lr} and A_{fb} ;
 Calculate the Cross-Entropy Loss, Minimum Category Confusion Loss, Loss of DADSL;
 Update parameter in the TSGCN model with Adam;
END for

3 Experiments and results

We conducted experiments with two datasets: SJTU Emotion EEG Dataset (SEED) and DEAP [55, 56].

3.1 Datasets

The SEED dataset contains emotional EEG data recorded from fifteen Chinese subjects (eight females and seven males) while watching movie clips. The EEG signals were recorded from 62 channels with a sampling rate of 1000 Hz, and then the EEG signals were down-sampled to 200 Hz and band-pass filtered to 0–75 Hz. In the SEED dataset, movie clips elicited three kinds of emotion: positive, negative, and neutral. An experiment consisted of three sets of sessions, with each session containing fifteen trials. Each trial had a 5-s hint before the movie clip was presented. Then, participants were required to watch a 4-min long movie clip. After watching the movie clip, each participant participated in a 45-s self-assessment. Between two trials, there was a 15-s resting period.

The DEAP dataset contains EEG data from 32 subjects (sixteen females and sixteen males) while watching music videos. The EEG signals were recorded from 32 channels with a sampling rate of 512 Hz. In addition, eight peripheral signals were recorded. With DEAP, an experiment consisted of forty trials, with a short break beginning with the twentieth trial. In each trial, there was a 3-s presentation of a fixation cross for baseline recording. Then, participants watched a 1-min display of a pre-selected music video, followed by a nine-point self-assessment of the video regarding arousal, valence, and dominance. In

the preprocessing step, EEG signals were down-sampled to 128 Hz. Line noise, blinks, and other artifacts were removed.

We evaluated the performance of our TSGCN model based on these two datasets. The TSGCN model was used to predict three categories of emotion labels (positive, negative, and neutral) for the SEED dataset. According to Yin's study, the TSGCN model was used to predict two categories of emotions on the arousal and valence dimensions (high valence vs. low valence, high arousal vs. low arousal) for the DEAP dataset [57].

3.2 Preprocessing and model implementation

SEED dataset For the preprocessed EEG signals in the SEED dataset, we used the Fast Fourier Transform (FFT) to filter the continuous EEG signals into the following five frequency bands: Delta [1-3 Hz], Theta [3-8 Hz], Alpha [8-13 Hz], Beta [13-30 Hz], and Gamma [30-50 Hz]. We divided the continuous EEG signals into 4-s non-overlapping epochs. Finally, for each sample, we computed DE features for every second in each epoch from each EEG channel in the five frequency bands.

DEAP dataset For the DEAP dataset, we used the FFT method to filter the EEG data into the following four frequency bands: Theta [4-8 Hz], Alpha [8-13 Hz], Beta [13-30 Hz], and Gamma [30-45 Hz]. We segmented the filtered data into 60-s trials and removed a 3-s pre-trial baseline. Finally, we used a 1-s non-overlapping Hamming window to extract DE features from each EEG channel in the four frequency bands. Then, we combined three seconds of DE features into one sample.

From the data preprocessing, we obtained a total number of 842 samples for each subject in the SEED dataset and a total number of 800 samples for each subject in the DEAP dataset. To train our TSGCN model, we used gradient descent and Adam to optimize model parameters. The hyperparameters of TSGCN were selected as follows: the dropout rate of the fully connected layer was 0.5, the learning rate, η , was set at $1e-4$, and the batch size was set to 64. The number of layers for both GCN and LSTM is 2, and they are used to extract time and frequency domain information. d , d' , and d'' denote 4, 256, and 128, respectively. Other hyperparameters were adjusted according to the experiment.

To validate the performance of our TSGCN model, we used the following as benchmarks for comparison on the SEED dataset: Support Vector Machine (SVM) [58], Deep Belief Network (DBN) [59], Dynamical Graph Convolutional Neural Network (DGCNN) [32], Bi-hemisphere Domain Adversarial Neural Network (BiDANN) [51], and Regularized Graph Neural Network (RGNN) [34]. Additionally, as benchmarks for comparison on

Table 1 Subject-dependent classification accuracy (mean/std) on the SEED dataset

Model	ACC/STD(%)
SVM	83.99/9.92
DBN	86.08/8.34
DGCNN	90.40/8.49
BiDANN	92.38/7.04
RGNN	94.24/5.95
TSGCN	94.98/4.59

Table 2 Subject-dependent classification accuracy (mean/std) on the DEAP dataset

Model	Valence ACC/STD(%)	Arousal ACC/STD(%)
SVM	51.60/6.32	55.80/9.49
DBN	55.04/8.82	62.43/11.87
BiDANN	84.51/6.32	80.99/7.04
DGCNN	86.06/2.61	85.61/2.44
ATDD-LSTM	90.91/12.95	90.87/11.32
TSGCN	92.45/4.51	91.73/3.82

the DEAP dataset, we used the following: SVM, DBN, BiDANN, DGCNN, and Attention-based LSTM with Domain Discriminator (ATDD-LSTM) [60].

3.3 Validation of subject-dependent classification on SEED and DEAP datasets.

In the subject-dependent classification experiments for the SEED dataset, we evaluated our TSGCN model according to the previous experimental settings [34, 51]. In each session, EEG data from the first nine trials were used as training data, and the data from the remaining six trials were used as test data. Then, we calculated the average accuracy of the fifteen subjects. For the DEAP dataset, we evaluated the performance of our TSGCN model according to the experimental settings used by X. Du et al. [60]. Data from 32 EEG channels were used for emotion recognition; the eight peripheral channels were removed. For each subject, EEG data from the first 39 trials were used as training data, and the data from the remaining trials were used as test data. This process was repeated until all the trials were used once to collect test data. Average accuracy was derived by averaging 32 subjects.

The subject-dependent classification accuracy of TSGCN on SEED and DEAP datasets is shown in Table 1 and Table 2, respectively. TSGCN achieves the highest accuracy among all models in both datasets. Compared with previous studies, TSGCN considers both emotional

features from multi-domain and noisy labels to improve robustness.

To validate the performance of TSGCN on each category of emotion, we constructed confusion matrices of TSGCN, as shown in Fig. 3a, b, and c. For the SEED dataset, TSGCN performs well in recognizing neutral and positive emotion labels. However, the accuracy of negative emotion label recognition is poor. Movie clips with negative emotion labels are discriminated against easily as neutral. For the DEAP dataset, both in the Valence and Arousal labels, TSGCN performs better in recognition of the High class than the Low class.

3.4 Validation of subject-independent classification on SEED and DEAP datasets

In the subject-independent classification experiments, for both SEED and DEAP datasets, based on previous experimental settings, we used the Leave-One-Out Cross Validation strategy to evaluate our TSGCN model.

The subject-independent classification accuracy of TSGCN on the SEED and DEAP datasets is shown in Tables 3 and 4, respectively. TSGCN achieves the highest accuracy among all models on the SEED dataset. For the DEAP dataset, recognition of Valence labels has the highest accuracy among all models. Recognition of Arousal labels achieves a comparable result with the latest studies.

The confusion matrices of subject-independent classification are shown in Fig. 4a, b and c. For the SEED dataset, TSGCN performs well in identifying positive emotion labels. However, the accuracy of both neutral and negative emotion labels is lower than that of positive emotion labels. This result indicates that neutral and negative emotion labels are more difficult to detect and easily confused with each other for the subject-independent task. For the DEAP dataset, the accuracy is much higher with the High-class recognition than that of the Low-class. In the dimensional emotion model, both Low Valence and Low Arousal are biased toward negative emotions, indicating that the recognition results of TSGCN for negative emotions are not ideal.

3.5 Ablation study

We performed an ablation study on each module of TSGCN and each connectivity matrix in the DGCNN module to verify their effectiveness in TSGCN. The results of the ablation experiments are shown in Tables 5 and 6, respectively. Table V shows the results of each module in TSGCN for subject-dependent and subject-independent classifications on the SEED dataset.

Subject-dependent In subject-dependent classification, we compared the performance of each module in TSGCN by removing this module. TSGCN model that removes DGCNN and retains BiLSTM is designated

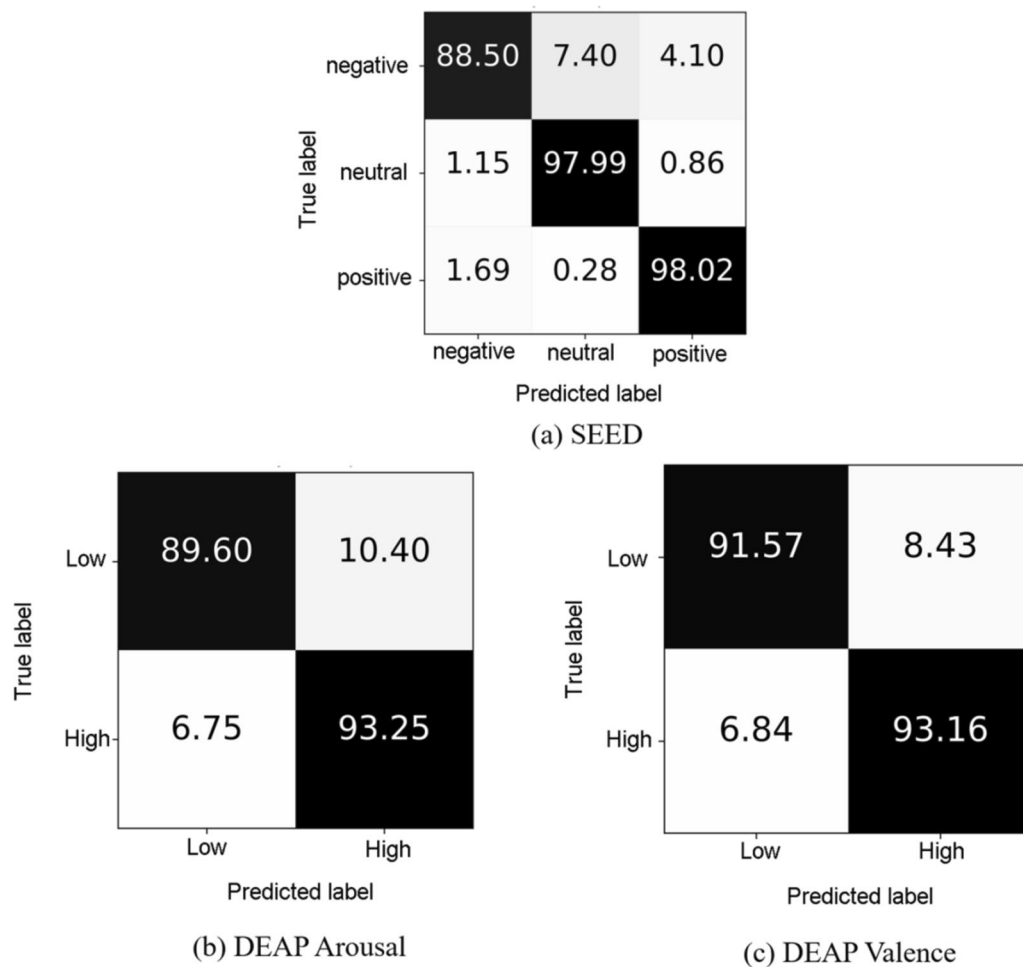


Fig. 3 Confusion matrices of subject-dependent classification. **(a)** The confusion matrix for the three categories of emotion labels on the SEED dataset. **(b)** The confusion matrix for the high and low Arousal labels on the DEAP dataset. **(c)** The confusion matrix for the high and low Valence labels on the DEAP dataset

Table 3 Subject-independent classification accuracy (mean/std) on the SEED dataset

Model	ACC/STD(%)
SVM	56.73/16.29
DBN	58.50/10.94
DGCNN	79.95/9.02
BiDANN	84.14/6.87
RGNN	85.30/6.72
TSGCN	86.13/5.29

Table 4 Subject-independent classification accuracy (mean/std) on the DEAP dataset

Model	Valence ACC/STD(%)	Arousal ACC/STD(%)
SVM	48.58/4.24	50.75/4.87
DBN	51.64/7.84	56.68/13.28
BiDANN	61.01/5.65	65.12/9.37
DGCNN	59.29/6.83	61.10/12.28
ATDD-LSTM	69.06/6.37	72.97/6.57
TSGCN	70.65/4.19	72.54/6.52

by—DGCNN. TSGCN model that removes BiLSTM and retains DGCNN is designated by -BiLSTM. TSGCN model that removes the MCC loss function is designated by -MCC. In Table 5, it is shown that, in the SEED dataset,—DGCNN has higher classification accuracy than -BiLSTM, which indicates that

time–frequency information is more important than spatial information for emotion recognition. Furthermore, MCC improves the performance of TSGCN by about 3%, demonstrating that reducing interference of noise labels enhances the performance of emotion recognition.

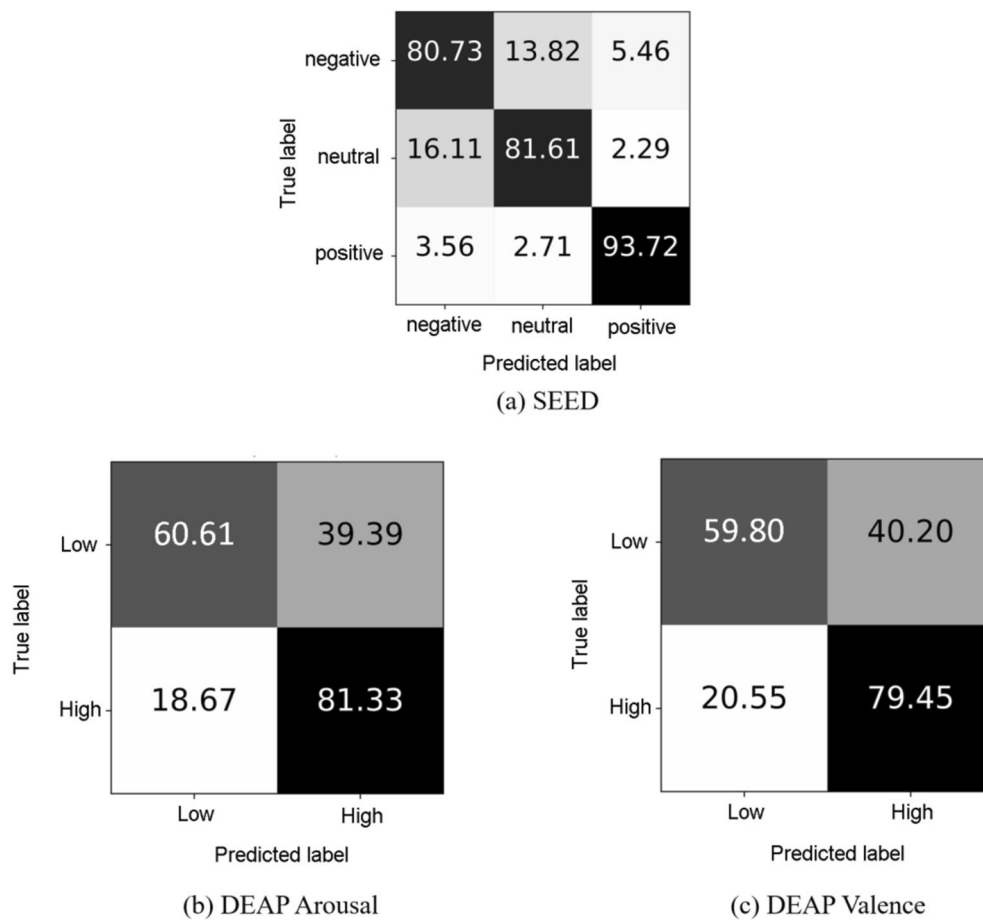


Fig. 4 Confusion matrices of subject-independent classification. (a) The confusion matrix for the three categories of emotion labels on the SEED dataset. (b) The confusion matrix for the high and low Arousal labels on the DEAP dataset. (c) The confusion matrix for the high and low Valence labels on the DEAP dataset

Table 5 Ablation studies on the SEED dataset

Model	ACC/STD(%)
TSGCN	94.98/5.59
-BiLSTM	86.95/8.76
-DGCNN	90.20/7.72
-MCC	92.07/6.98
DGL_N+DGL_TF	86.13/5.29
-DGL_TF	84.11/7.27
-DGL_N	82.68/8.11

‘-’ indicates the following component is removed

Subject-independent In Subject-Independent experiments, we verified the effectiveness of the Deep-Shallow feature Dynamic Adversarial Learning (DSDAL) and compared it with different adversarial learning modules. TSGCN model implemented with DSDAL is represented by DGL_N+DGL_TF. TSGCN model where shallow feature adversarial learning is used is represented by

Table 6 Ablation studies on different connectivity matrices on the SEED dataset

Connectivity	ACC/STD(%)
(1) Full Connectivity	90.23/5.35
(2) Up-down Connectivity	89.74/6.56
(3) Left-right Connectivity	89.59/6.87
(2) + (3)	91.88/6.29
(1) + (2) + (3)	94.98/4.59

DGL_TF. TSGCN model in which deep feature adversarial learning is used is indicated by DGL_N.

As seen in Table 5, -DGL_TF performs better than -DGL_N because the shallow features can be adjusted while the deep features of the graph network are aligned. In addition, compared with other methods, DSDAL significantly improves the accuracy of subject-independent classification, which proves that aligning the features of both deep and shallow layers improves

the accuracy of cross-subject emotion recognition. In DSDAL, the shallow features are first aligned; then, the deep features are adjusted and aligned. By applying this strategy, the influence of shallow feature distribution on deep feature distribution is reduced.

Table 6 shows the results of different connectivity matrices in the DGCNN module for subject-dependent classifications on the SEED dataset. It is shown that the DGCNN module with the up-down connectivity matrix has a comparative accuracy with the left-right connectivity matrix. The DGCNN module with a full connectivity matrix performs better than the up-down and left-right connectivity matrix. DGCNN module with the combination of up-down and left-right connectivity matrices performs better than the full connectivity matrix. Among them, the DGCNN module, which combines three connectivity matrices, has the best performance. These results indicate that up-down and left-right connectivity matrices can provide complementary information on the topological relationship between electrode channels to improve recognition accuracy.

4 Discussion

4.1 Visualization of channel relations

We visualized weights of the learnable connectivity matrix to study the topological relationship between channels across the brain scalp. The visualization is as follows: we first normalized the learnable connectivity matrix, A_{all} , and plotted it as a heatmap. As shown in Fig. 5a, increased connection weight is indicated by a darker color in the matrix. We found that, although there are obvious connections between local adjacent channels, there is no obvious connection between channels that are far apart. In addition, connections between different brain regions were visualized according to the connectivity matrix. The results are shown in Fig. 5b. We found tight connections between local channels in different brain regions. In contrast, there are relatively sparse connections between different brain regions, which is in line with the small-world characteristics of the brain network [61]. Among these brain regions, the connectivity between parietal and occipital regions is relatively tight, whereas there are relatively sparse long-range connections between prefrontal-central and central-parietal regions, indicating the prefrontal regions are involved in the formation of high-level emotional reactions [62].

4.2 Influence of subjective rating

We performed additional analysis on our experiments to further investigate the relationship between subjective ratings and emotion recognition accuracy. MCC loss is designed to capture the discrepancy between a subjective rating and predefined labels. We performed a

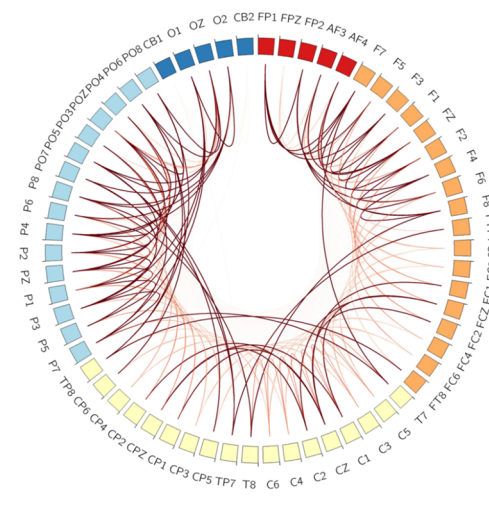
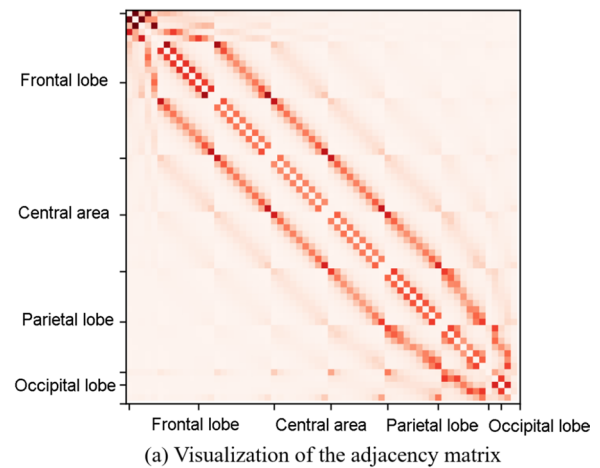
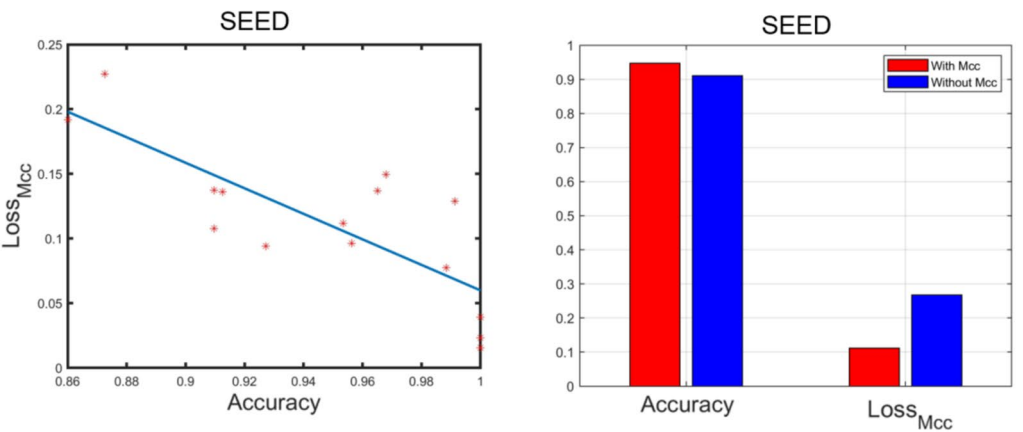
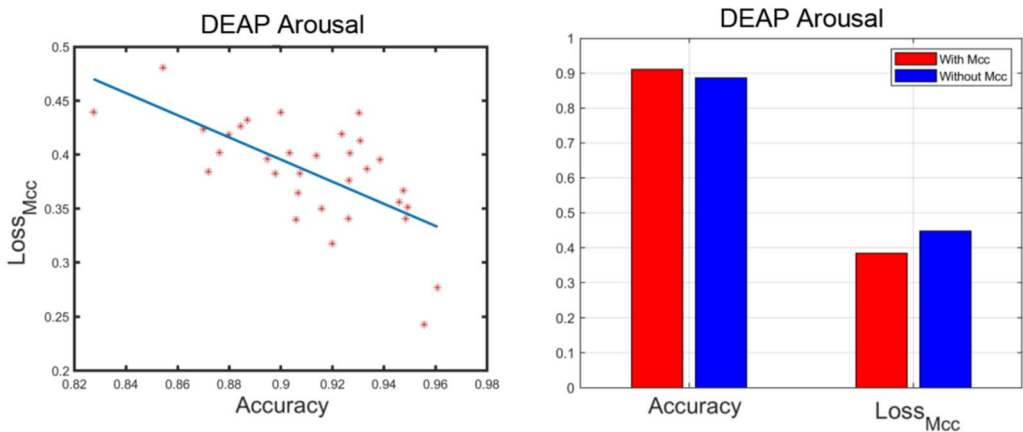


Fig. 5 Topological relationship of the learnable connection matrix. **(a)** The adjacency matrix between channels in the frontal, central, parietal, and occipital brain regions. A higher connection weight is indicated by a darker color in the matrix. **(b)** The connectivity between channels in different brain regions. All of the channels are plotted into a circle, and connections between channels are plotted by red lines. A higher connection weight is indicated by a darker line color

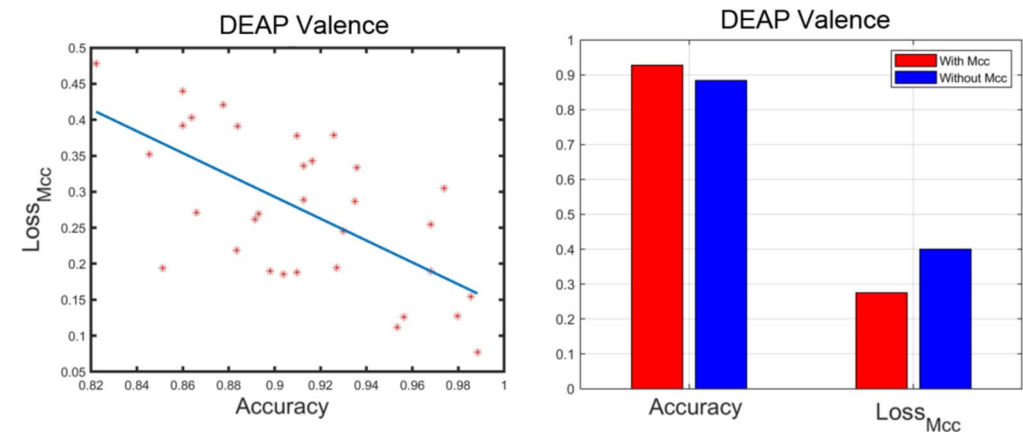
correlation analysis between MCC loss and recognition accuracy on the SEED and DEAP datasets. As shown in Fig. 6, there is a significant correlation between MCC loss and recognition accuracy in both datasets ($r = 0.63$, $p < 0.01$ for the SEED dataset, $r = 0.414$, $p < 0.01$ for the Valence label of the DEAP dataset, $r = 0.432$, $p < 0.01$ for the Arousal label of the DEAP dataset). When MCC loss increases, recognition accuracy decreases. These results verify that subjective ratings play an essential role in emotion recognition. Inconsistencies between



(a) Correlation analysis between MCC loss and accuracy on the SEED dataset.



(b) Correlation analysis between MCC loss and accuracy on Arousal labels of the DEAP dataset..



(c) Correlation analysis between MCC loss and accuracy on Valence labels of the DEAP dataset.

Fig. 6 (See legend on previous page.)

(See figure on next page.)

Fig. 6 Relationship between subjective ratings and emotion recognition accuracy. **(a)** The accuracy is negatively correlated with MCC loss on the SEED dataset. The classification accuracy increases, and the MCC loss decreases after implementing the MCC loss function. **(b)** The accuracy is negatively correlated with MCC loss on Arousal labels of the DEAP dataset. The classification accuracy on Arousal labels increases, and the MCC loss decreases after implementing the MCC loss function. **(c)** The accuracy is negatively correlated with MCC loss on Valence labels of the DEAP dataset. The classification accuracy on Valence labels increases, and the MCC loss decreases after implementing the MCC loss function

Table 7 T-test analysis of MCC loss before and after using the MCC loss function

Dataset	With MCC	Without MCC	<i>p</i>
SEED	0.1116/0.0582	0.2670/0.2093	0.007406
DEAP-arousal	0.3839/0.0489	0.4473/0.1526	0.002479
DEAP-valence	0.2745/0.1055	0.3998/0.2027	0.001606

Table 8 T-test analysis of ACC before and after using the MCC loss function

Dataset	With MCC	Without MCC	<i>p</i>
SEED	94.98/5.59	92.07/6.98	0.162886
DEAP-arousal	91.73/3.82	88.80/5.72	0.055633
DEAP-valence	92.45/4.51	88.38/5.1	0.000241

subjective ratings and predefined labels can impact classification accuracy.

To verify the effectiveness of the proposed MCC in eliminating the influence of subjective ratings, we statistically compared recognition accuracy and MCC loss before and after using the MCC loss function. A T-test was carried out for statistical analysis. As shown in Fig. 6, after implementing the MCC loss function in our model (p-value for SEED and DEAP datasets are shown in Tables 7 and 8, respectively), accuracy significantly increases, and MCC loss significantly decreases. Thus, our MCC loss function effectively reduces the influence caused by inconsistencies of subjective ratings.

4.3 Effectiveness of DSDAL algorithm

To further verify that DSDAL reduces the influences of mismatched domains across subjects and improves recognition performance, we visualized feature distribution with four subjects in the SEED dataset using the t-SNE method. Figure 7 shows the visualization results of discriminative features the Temporal-Spectral Graph Convolution Network learned on the subject-independent classification of the SEED dataset using DGL_TF, DGL_N, and DSDAL. In Fig. 7, the raw EEG features of positive, negative, and neutral emotion labels are mixed and almost

unrecognizable. After training with the three methods, the three emotion labels are well distinguished, and positive emotion labels are identified more easily than neutral and negative emotion labels. In addition, in subjects 1 and 3, it can be seen that DSDAL has better separability than DGL-TF and DGL-N, which shows that DSDAL is more advantageous than the other two methods.

5 Conclusion

We proposed a novel TSGCN model to deal with the challenges in EEG emotion recognition. Extensive experiments were conducted on the SEED and DEAP datasets to validate the performance of our method. Results demonstrate that, on these datasets, our proposed TSGCN method achieves better performance than the state-of-the-art methods. Detailed analysis of each module provides insights into the superiority of our proposed method. The better performance of TSGCN is attributed mainly to the fact that TSGCN not only uses discrete information from time, spatial, and frequency domains to improve recognition performance but also minimizes the effect of noisy labels caused by subjective ratings. We also proposed the Deep and Shallow feature Dynamic

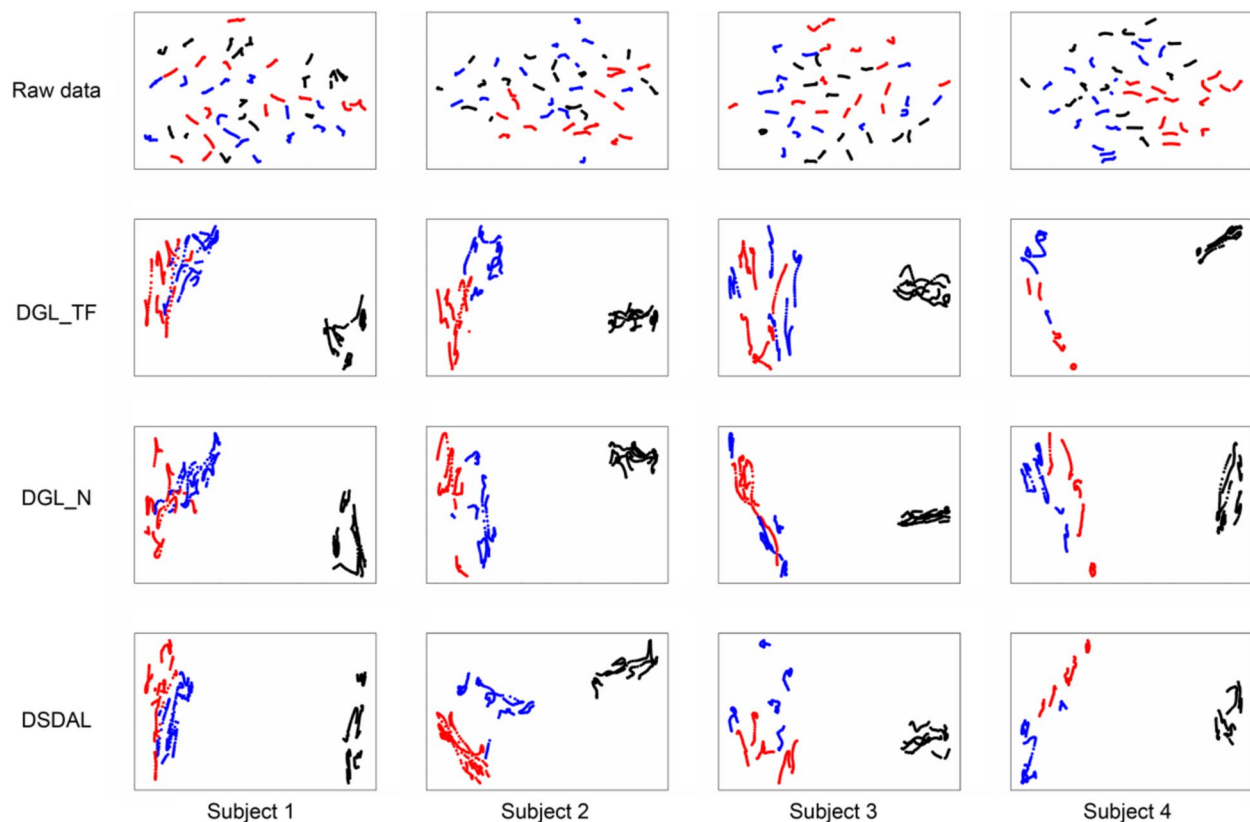


Fig. 7 T-SNE visualization results for different methods on the subject-independent classification of SEED dataset. The red represents negative emotion labels, the blue represents neutral emotion labels, and the black represents positive emotion labels

Adversarial Learning and integrated it with TSGCN to deal with the challenge of cross-subject emotion recognition. Experimental results confirm that TSGCN with DSDAL performs better than the original TSGCN in terms of subject-independent classification.

Abbreviations

ATDD-LSTM	Attention-based LSTM with domain discriminator
BiDANN	Bi-hemispheres domain adversarial neural network
BiLSTM	Bi-directional long short-term memory
CCE	Categorical cross entropy
CNN	Convolutional neural network
DANN	Domain adversarial neural network
DBN	Deep belief network
DE	Diffusion entropy
DGCNN	Dynamic graph convolutional neural network
DSDAL	Deep and shallow feature dynamic adversarial learning
EEG	Electroencephalogram
FFT	Fast fourier transform
fMRI	Functional magnetic resonance imaging
GNN	Graph neural network
GRL	Gradient reversal layer
LSTM	Long short-term memory
MCC	Minimum category confusion
MMD	Maximum mean discrepancy
PCRN	Parallel convolutional recurrent neural network
PSD	Power spectral density
RGNN	Regularized graph neural network
RNN	Recurrent neural networks

SEED	SJTU emotion EEG dataset
ST-SCGNN	Spatio-temporal self-constructing graph neural network
SVM	Support vector machine
TSGCN	Temporal-spectral graph convolutional network
WT	Wavelet transform

Acknowledgements

Not applicable.

Author contributions

RL: Conceptualization, methodology, writing—original draft preparation, writing—review editing, and project administration. XWY: Methodology, data curation, validation, writing—review editing. JL: Data curation, methodology, and validation. JSZ: Conceptualization, methodology, validation, writing—review editing, funding acquisition, and supervision. All authors contributed to the article and approved the submitted version.

Funding

This work was supported by the National Natural Science Foundation of China (61772440, 62007016), the Collaborative Innovation Platform Project for Fuzhou-Xiamen-Quanzhou National Independent Innovation Demonstration Zone (No. 2022-P-028), the Project of Industry-university-research Collaborative Innovation in Fujian Province Universities (No. 2022Y4004), the 2022 Hulunbuir City's Special Key Project for Revitalizing City by Science and Technology (No. 2022HZZX002).

Availability of data and materials

This experiment uses the public SEED and DEAP EEG datasets with emotional labels. The DEAP dataset can be accessed online at <http://www.eecs.qmul.ac>.

uk/, and the SEED dataset can be accessed online at <https://bcmi.sjtu.edu.cn/~seed/index.html>.

Declarations

Ethics approval and consent to participate

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 7 September 2024 Accepted: 12 November 2024

Published online: 18 December 2024

References

- Picard RW (2000) Affective computing. Springer, Berlin Heidelberg, Berlin
- Wells AE, Hunnkin LM, Ash DP, Van Goozen SH (2021) Improving emotion recognition is associated with subsequent mental health and well-being in children with severe behavioural problems. *Eur Child Adolesc Psychiatry* 30(11):1769–1777. <https://doi.org/10.1007/s00787-020-01652-y>
- Mumenthaler C, Sander D, Manstead AS (2018) Emotion recognition in simulated social interactions. *IEEE Trans Affect Comput* 11(2):308–312. <https://doi.org/10.1109/TAFFC.2018.2799593>
- Cowie R, Douglas-Cowie E, Tsapatsoulis N, Votsis G, Kollias S, Fellenz W, Taylor JG (2001) Emotion recognition in human-computer interaction. *IEEE Signal Proc Mag* 18(1):32–80. <https://doi.org/10.1109/79.911197>
- Plutchik R (1980) A general psychoevolutionary theory of emotion. *Emotion Theory Res Exper*. <https://doi.org/10.1016/B978-0-12-558701-3.50007-7>
- Mehrabian A (1996) Pleasure-arousal-dominance: a general framework for describing and measuring individual differences in temperament. *Curr Psychol* 14:261–292. <https://doi.org/10.1007/BF02686918>
- Sebe N, Cohen I, Gevers T (2005) Huang TS Multimodal approaches for emotion recognition: a survey. *Internet Imaging*. <https://doi.org/10.1117/12.600746>
- Jiang Y, Li W, Hossain MS, Chen M, Alelaiwi A, Al-Hammadi M (2020) A snapshot research and implementation of multimodal information fusion for data-driven emotion recognition. *Information Fusion* 53:209–221. <https://doi.org/10.1016/j.inffus.2019.06.019>
- He Z, Zhong Y, Pan J (2022) An adversarial discriminative temporal convolutional network for EEG-based cross-domain emotion recognition. *Comput Biol Med* 141:105048. <https://doi.org/10.1016/j.combiomed.2021.105048>
- Alarcao SM, Fonseca MJ (2017) Emotions recognition using EEG signals: a survey. *IEEE Trans Affect Comput* 10(3):374–393. <https://doi.org/10.1109/TAFFC.2017.2714671>
- Urigüen JA, García-Zapirain B (2015) EEG artifact removal—state-of-the-art and guidelines. *J Neural Eng* 12(3):031001. <https://doi.org/10.1088/1741-2560/12/3/031001>
- Pessoa L (2008) On the relationship between emotion and cognition. *Nat Rev Neurosci* 9(2):148–158. <https://doi.org/10.1038/nrn2317>
- Esslen M, Pascual-Marqui RD, Hell D, Kochi K, Lehmann D (2004) Brain areas and time course of emotional processing. *Neuroimage* 21(4):1189–1203. <https://doi.org/10.1016/j.neuroimage.2003.10.001>
- Li J, Zhang Z, He H (2018) Hierarchical convolutional neural networks for EEG-based emotion recognition. *Cogn Comput* 10:368–380. <https://doi.org/10.1007/s12559-017-9533-x>
- Phan KL, Taylor SF, Welsh RC, Ho S-H, Britton JC, Liberzon I (2004) Neural correlates of individual ratings of emotional salience: a trial-related fMRI study. *Neuroimage* 21(2):768–780. <https://doi.org/10.1016/j.neuroimage.2003.09.072>
- Hamann S, Canli T (2004) Individual differences in emotion processing. *Curr Opin Neurobiol* 14(2):233–238. <https://doi.org/10.1016/j.conb.2004.03.010>
- Zheng W-L, Zhang Y-Q, Zhu J-Y, Lu B-L (2015) Transfer components between subjects for EEG-based emotion recognition. In: 2015 international conference on affective computing and intelligent interaction (ACII). <https://doi.org/10.1109/ACII.2015.7344684>
- Pandey P, Seeja K (2022) Subject-independent emotion detection from EEG signals using deep neural network. *J King Saud Univ Comput Inform Sci*. <https://doi.org/10.1016/j.jksuci.2019.11.003>
- Pan SJ, Tsang IW, Kwok JT, Yang Q (2010) Domain adaptation via transfer component analysis. *IEEE Trans Neural Networks* 22(2):199–210. <https://doi.org/10.1109/TNN.2010.2091281>
- Wang X-W, Nie D, Lu B-L (2014) Emotional state classification from EEG data using machine learning approach. *Neurocomputing* 129:94–106. <https://doi.org/10.1016/j.neucom.2013.06.046>
- Alhagry S, Fahmy AA, El-Khoribi RA (2017) Emotion recognition based on EEG using LSTM recurrent neural network. *Int J Adv Comput Sci Appl*. <https://doi.org/10.14569/IJACSA.2017.081046>
- Ma J, Tang H, Zheng W-L, Lu B-L (2019) Emotion recognition using multimodal residual LSTM network. In: Proceedings of the 27th ACM international conference on multimedia. <https://doi.org/10.1145/3343031.3350871>
- Xiao G, Shi M, Ye M, Xu B, Chen Z, Ren Q (2022) 4D attention-based neural network for EEG emotion recognition. *Cognit Neurodynam*. <https://doi.org/10.1007/s11571-021-09751-5>
- Wen Z, Xu R, Du J (2017) A novel convolutional neural networks for emotion recognition based on EEG signal. In: 2017 International Conference on Security, Pattern Analysis, and Cybernetics (SPAC). <https://doi.org/10.1109/SPAC.2017.8304360>
- Jung T-P, Sejnowski TJ (2019) Utilizing deep learning towards multimodal bio-sensing and vision-based affective computing. *IEEE Trans Affect Comput* 13(1):96–107. <https://doi.org/10.1109/TAFFC.2019.2916015>
- Moon S-E, Jang S, Lee J-S (2018) Convolutional neural network approach for EEG-based emotion recognition using brain connectivity and its spatial information. In: 2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). <https://doi.org/10.48550/arXiv.1809.04208>
- Li T, Fu B, Wu Z, Liu Y (2023) Eeg-based emotion recognition using spatial-temporal-connective features via multi-scale CNN. *IEEE Access* 11:41859–41867. <https://doi.org/10.1109/ACCESS.2023.3270317>
- Yang Y, Wu Q, Qiu M, Wang Y, Chen X (2018) Emotion recognition from multi-channel EEG through parallel convolutional recurrent neural networks. In: 2018 international joint conference on neural networks (IJCNN). <https://doi.org/10.1109/IJCNN.2018.8489331>
- Zhang T, Zheng W, Cui Z, Zong Y, Li Y (2018) Spatial-temporal recurrent neural network for emotion recognition. *IEEE T Cybernetics* 49(3):839–847. <https://doi.org/10.1109/TCYB.2017.2788081>
- Hou J, Luo C, Qin F, Shao Y, Chen X (2023) FuS-GCN: Efficient B-rep based graph convolutional networks for 3D-CAD model classification and retrieval. *Adv Eng Inform* 56:102008. <https://doi.org/10.1016/j.aei.2023.102008>
- Jin M, Du C, He H, Cai T, Li J (2024) PGCN: Pyramidal graph convolutional network for EEG emotion recognition. *IEEE Trans Multimedia*. <https://doi.org/10.1109/TMM.2024.3385676>
- Song T, Zheng W, Song P, Cui Z (2018) EEG emotion recognition using dynamical graph convolutional neural networks. *IEEE Trans Affect Comput* 11(3):532–541. <https://doi.org/10.1109/TAFFC.2018.2817622>
- Pan J, Liang R, He Z, Li J, Liang Y, Zhou X, He Y, Li Y (2023) ST-SCGNN: a spatio-temporal self-constructing graph neural network for cross-subject EEG-based emotion recognition and consciousness detection. *IEEE J Biomed Health Inform*. <https://doi.org/10.1109/JBHI.2023.3335854>
- Zhong P, Wang D, Miao C (2020) EEG-based emotion recognition using regularized graph neural networks. *IEEE Trans Affect Comput* 13(3):1290–1301. <https://doi.org/10.1109/TAFFC.2020.2994159>
- Song T, Liu S, Zheng W, Zong Y, Cui Z, Li Y, Zhou X (2021) Variational instance-adaptive graph for EEG emotion recognition. *IEEE Trans Affect Comput* 14(1):343–356. <https://doi.org/10.1109/TAFFC.2021.3064940>
- Han B, Yao Q, Liu T, Niu G, Tsang IW, Kwok JT, Sugiyama M (2020) A survey of label-noise representation learning: Past, present and future. *arXiv preprint arXiv:201104406*. <https://doi.org/10.48550/arXiv.2011.04406>
- Sukhbaatar S, Bruna J, Paluri M, Bourdev L, Fergus R (2014) Training convolutional networks with noisy labels. *arXiv preprint arXiv:1406.2080*. <https://doi.org/10.48550/arXiv.1406.2080>

38. Patrini G, Rozza A, Krishna Menon A, Nock R, Qu L (2017) Making deep neural networks robust to label noise: a loss correction approach. *Proc IEEE Conf Comput Vision Pattern Recog*. <https://doi.org/10.48550/arXiv.1609.03683>
39. Azadi S, Feng J, Jegelka S, Darrell T (2015) Auxiliary image regularization for deep cnns with noisy labels. *arXiv preprint arXiv:1511.07069*. <https://doi.org/10.48550/arXiv.1511.07069>
40. Berthelot D, Carlini N, Goodfellow I, Papernot N, Oliver A, Raffel CA (2019) Mixmatch: a holistic approach to semi-supervised learning. *Adv Neural Inform Proc Syst*. <https://doi.org/10.48550/arXiv.1905.02249>
41. Zhang Z, Sabuncu M (2018) Generalized cross entropy loss for training deep neural networks with noisy labels. *Adv Neural Inform Proc Syst*. <https://doi.org/10.48550/arXiv.1805.07836>
42. Jiang L, Zhou Z, Leung T, Li L-J, Fei-Fei L (2018) Mentornet: learning data-driven curriculum for very deep neural networks on corrupted labels. *Int Conf Mach Learn*. <https://doi.org/10.48550/arXiv.1712.05055>
43. Han B, Yao Q, Yu X, Niu G, Xu M, Hu W, Tsang I, Sugiyama M (2018) Co-teaching: robust training of deep neural networks with extremely noisy labels. *Adv Neural Inform Proc Syst*. <https://doi.org/10.48550/arXiv.1804.06872>
44. Quan J, Li Y, Wang L, He R, Yang S, Guo L (2023) EEG-based cross-subject emotion recognition using multi-source domain transfer learning. *Biomed Signal Process Control* 84:104741. <https://doi.org/10.1016/j.bspc.2023.104741>
45. Tzeng E, Hoffman J, Zhang N, Saenko K, Darrell T (2014) Deep domain confusion: Maximizing for domain invariance. *arXiv preprint arXiv:1412.3474*. <https://doi.org/10.48550/arXiv.1412.3474>
46. Long M, Cao Y, Wang J, Jordan M (2015) Learning transferable features with deep adaptation networks. *Int Conf Mach Learn*. <https://doi.org/10.48550/arXiv.1502.02791>
47. Sun B, Saenko K (2016) Deep coral: Correlation alignment for deep domain adaptation. In: Hua G (ed) *Computer Vision—ECCV 2016 Workshops Amsterdam The Netherlands October 8–10 and 15–16 Proceedings*. Springer, Cham, pp 443–450
48. Li H, Jin Y-M, Zheng W-L, Lu B-L (2018) Cross-subject emotion recognition using deep adaptation networks. In: Long Cheng (eds). *Neural Information Processing: 25th International Conference, ICONIP 2018, Siem Reap, Cambodia, December 13–16, 2018, Proceedings, Part V*. Springer: Cham. pp 403–413
49. Zhang X, Liang W, Ding T, Pan J, Shen J, Huang X (2019) Individual similarity guided transfer modeling for EEG-based emotion recognition. *Int Conf Bioinform Biomed*. <https://doi.org/10.1109/BIBM47256.2019.8982972>
50. Ganin Y, Ustinova E, Ajakan H, Germain P, Larochelle H, Laviolette F, March M, Lempitsky V (2016) Domain-adversarial training of neural networks. *J Mach Learn Res* 17(59):1–35. <https://doi.org/10.48550/arXiv.1505.07818>
51. Li Y, Zheng W, Zong Y, Cui Z, Zhang T, Zhou X (2018) A bi-hemisphere domain adversarial neural network model for EEG emotion recognition. *IEEE Trans Affect Comput* 12(2):494–504. <https://doi.org/10.1109/TAFFC.2018.2885474>
52. Achard S, Bullmore E (2007) Efficiency and cost of economical brain functional networks. *PLoS Comput Biol* 3(2):e17. <https://doi.org/10.1371/journal.pcbi.0030017>
53. Jin Y, Wang X, Long M, Wang J (2020) Minimum class confusion for versatile domain adaptation. In: Vedaldi A (ed) *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXI*. Springer, Cham, pp 464–480
54. Yu C, Wang J, Chen Y (2019) Transfer learning with dynamic adversarial adaptation network. *Int Conf Data Mining (ICDM)*. <https://doi.org/10.1109/ICDM.2019.00088>
55. Shi L-C, Jiao Y-Y (2013) Differential entropy feature for EEG-based vigilance estimation. *Ann Int Conf IEEE Eng Med Biol Soc (EMBC)*. <https://doi.org/10.1109/EMBC.2013.6611075>
56. Koelstra S, Muhl C, Soleymani M, Lee J-S, Yazdani A, Ebrahimi T, Pun T, Nijholt A, Patras I (2011) Deap: a database for emotion analysis; using physiological signals. *IEEE Trans Affect Comput* 3(1):18–31. <https://doi.org/10.1109/T-AFFC.2011.15>
57. Yin Z, Zhao M, Wang Y, Yang J, Zhang J (2017) Recognition of emotions using multimodal physiological signals and an ensemble deep learning model. *Comput Methods Programs Biomed* 140:93–110. <https://doi.org/10.1016/j.cmpb.2016.12.005>
58. Suykens JA, Vandewalle J (1999) Least squares support vector machine classifiers. *Neural Process Lett* 9:293–300. <https://doi.org/10.1023/A:1018628609742>
59. Zheng W-L, Lu B-L (2015) Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks. *IEEE Trans Auton Ment Dev* 7(3):162–175. <https://doi.org/10.1109/TAMD.2015.2431497>
60. Du X, Ma C, Zhang G, Li J, Lai Y-K, Zhao G, Deng X, Liu Y-J, Wang H (2020) An efficient LSTM network for emotion recognition from multichannel EEG signals. *IEEE Trans Affect Comput* 13(3):1528–1540. <https://doi.org/10.1109/TAFFC.2020.3013711>
61. Bullmore E, Sporns O (2012) The economy of brain network organization. *Nat Rev Neurosci* 13(5):336–349. <https://doi.org/10.1038/nrn3214>
62. Fesenko Z, Ptukha M, da Silva MM, de Carvalho RS, Tsytarev V, Gainetdinov RR, Faber J, Volnova AB (2024) Electrophysiological and behavioral markers of hyperdopaminergia in DAT-KO Rats. *Biomedicine* 12(9):2114. <https://doi.org/10.3390/biomedicine12092114>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.